EXERCISE-I

UNIT AND DIMENSIONS

1. Choose the option, whose pair doesn't have same dimensions.

(A) (Pressure × Volume) & Work done
(B) (Force × Time) & Change in momentum
(C) Kilocalorie & Joule
(D) Angle & mole

2. Which of the following can be a set of fundamental quantities

(A) length, velocity, time
(B) momentum, mass, velocity
(C) force, mass, velocity
(D) momentum, time, frequency

3. In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and unit of length is 10 m. In this system, one unit of power will correspond to

(A) 16 watts
(B) $\frac{1}{16}$ watts
(C) 25 watts
(D) none of these

4. The pressure of 10^6 dyne/cm^2 is equivalent to

(A) 10^5 N/m^2
(B) 10^6 N/m^2
(C) 10^7 N/m^2
(D) 10^8 N/m^2

5. The time dependence of a physical quantity p is given by $p = p_0 e^{(-\alpha t^2)}$ where α is constant and t is time. The constant α

(A) is dimensionless
(B) has dimensions T^{-2}
(C) has dimensions T^2
(D) has dimensions of p

6. Speed of gas molecules is given by the formula $v = \sqrt{\frac{3RT}{M}}$, there R is the universal gas constant (which appears in the equation PV = nRT) and T is temparature. Dimensions of M are -

(A) $M^1L^0T^0$
(B) M^1K^{-1}
(C) M^1L^{-1}
(D) M^1 mol^{-1}

7. **Statement-1 :** Method of dimensions cannot tell whether an equation is correct.

Statement-2 : A dimensionally incorrect equation may be correct.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

8. Considering Force (F), Velocity (V) and Energy (E) as fundamental quantities, match the correct dimensions of following quantities.

	Column-I		**Column-II**
(A)	Mass	(P)	$[F^{-1} V^{\circ} E^{1}]$
(B)	Light year	(Q)	$[F^{1} V^{1} E^{-1}]$
(C)	Frequency (1/T)	(R)	$[F^{3} V^{\circ} E^{-2}]$
(D)	Pressure	(S)	$[F^{\circ} V^{-2} E^{1}]$

9. The time period T of a physical pendulum depends on its weight ,W (which is a force), its moment of inertia I ([I] = [ML^2]) and distance of its centre from point of suspension *l*. If $T = KW^a I^b l^c$, where K is dimensionless constant. Find a, b, c.

10. (a) Suppose the speed of bullet is 1000 feet per second. Using 1 mile $\approx$ 5000 feet, what is the speed in miles per hour?

(b) A piece of aluminum foil has a mass per unit area equal to 0.01 g/cm^2. What is its mass per unit area in kg/m^2.

11. Potential energy of an object at any position x is given by the following relation $U = \dfrac{Bx^2}{A+\sqrt{x}}$. Find out dimensions of A and B.

12. The time period (T) of a spring mass system depends upon mass (m) & spring constant (k) & length of the spring (*l*) [$k = \dfrac{\text{Force}}{\text{length}}$]. Find the relation among, (T), (m), (*l*) & (k) using dimensional method.

13. The distance moved by a particle in time t from centre of a ring under the influence of its gravity is given by $x = a \sin\omega t$ where a & ω are constants. If ω is found to depend on the radius of the ring (r), its mass (m) and universal gravitational constant (G), find using dimensional analysis an expression for ω in terms of r, m and G.

14. If the velocity of light c, Gravitational constant G & Plank's constant h be chosen as fundamental units, find the dimension of mass, length & time in the new system.

15. A satellite is orbiting around a planet. Its orbital velocity (v_0) is found to depend upon

(a) Radius of orbit (R)

(b) Mass of planet (M)

(c) Universal gravitation constant (G)

Using dimensional analysis find an expression relating orbital velocity (v_0) to the above physical quantities.

VECTORS

16. **Statement-1 :** Unit vector has a unit though its magnitude is one.

Statement-2 : Unit vector is obtained by dividing a vector by its own magnitude.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

17. Let $\vec{A} = \hat{i} + \hat{j} + \hat{k}$, $\vec{B} = a\hat{i} - \hat{j} - \hat{k}$ and $\vec{C} = \hat{i} + \hat{j} + b\hat{k}$, given $\vec{A} \perp \vec{B}$ and $\vec{A} \perp \vec{C}$. Find angle between $\vec{B}$ & $\vec{C}$.

(A) 60° (B) 90° (C) 120° (D) 30°

18. Given $\vec{a} + \vec{b} + \vec{c} + \vec{d} = \vec{0}$, which of the following statements is/are correct :

(A) $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ must each be a zero vector

(B) The magnitude of $(\vec{a} + \vec{c})$ equals the magnitude of $(\vec{b} + \vec{d})$.

(C) The magnitude of $\vec{a}$ can never be greater than the sum of the magnitudes of $\vec{b}, \vec{c}$ and $\vec{d}$.

(D) $\vec{b} + \vec{c}$ must lie in the plane of $\vec{a}$ and $\vec{d}$ if $\vec{a}$ and $\vec{d}$ are not collinear.

19. If $\vec{A}.\vec{B} = \vec{C}.\vec{B}$, which of the following can not be a possible case :

(A) $\vec{A} = \vec{C}$

(B) angle between $\vec{A}$ and $\vec{B}$ = 30° and angle between $\vec{B}$ and $\vec{C}$ = 150° all three vectors are of equal magnitude.

(C) $\vec{B}$ is a null vector

(D) $\vec{A}, \vec{B}$ and $\vec{C}$ are along x, y and z-axis respectively.

20. **Statement-1 :** When two non-parallel forces $\vec{F}_1$ and $\vec{F}_2$ act on a body, the magnitude of the resultant force acting on the body is less than sum of F_1 and F_2.

Statement -2 : In a triangle, any side is less than the sum of the other two sides.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false.

21. Let $\vec{u}$ be a constant vector and $\vec{v}$ be a vector of constant magnitude such that $|\vec{v}| = 1/2\,|\vec{u}|$ and $|\vec{u}| \neq 0$.

Then the maximum possible angle between $\vec{u}$ and $\vec{u} + \vec{v}$ is :

(A) 60º (B) 120º (C) 30º (D) 150º

22. A man rows a boat with a speed of 18km/hr in northwest direction. The shoreline makes an angle of 15° south of west. Obtain the component of the velocity of the boat along the shoreline.

(A) 9 km/hr (B) $18\frac{\sqrt{3}}{2}$ km/hr (C) 18 cos15°km/hr (D) 18 cos75° km/hr

23. A bird moves from point $(1, -2)$ to $(4, 2)$. If the speed of the bird is 10 m/sec, then the velocity vector of the bird is :

(A) $5(\hat{i} - 2\hat{j})$ (B) $5(4\hat{i} + 2\hat{j})$ (C) $0.6\hat{i}+0.8\hat{j}$ (D) $6\hat{i}+8\hat{j}$

24. Two forces whose magnitudes are in the ratio 3 : 5 give a resultant of 28 N. If the angle of their inclination is 60º, find the magnitude of stronger force.

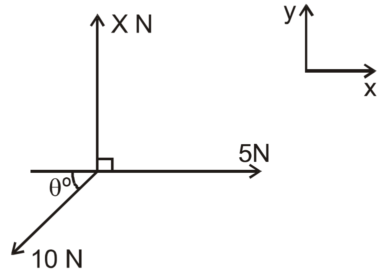
(A) 15 N (B) 12 N (C) 20 N (D) 18 N

25. If the angle between the unit vectors $\hat{a}$ and $\hat{b}$ is 60°, then $|\hat{a} - \hat{b}|$ is

(A) 0 (B) 1 (C) 2 (D) 4

Paragraph for Question Nos. 26 to 28

The diagram shows three coplanar forces acting on a particle P, which is in equilibrium.



26. What is the value of force X ?

(A) 5 (B) $5\sqrt{3}$

(C) 15 (D) None of these

27. What is the value of θ ?

(A) 30° (B) 37°

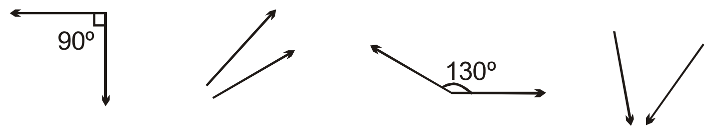
(C) 45° (D) 60°

28. The vector form for the force of 10N is

(A) $10\hat{i}$ (B) $5\hat{i}+5\sqrt{3}\hat{j}$

(C) $-5\hat{i}+5\sqrt{3}\hat{j}$ (D) $-5\hat{i}-5\sqrt{3}\hat{j}$

29. There are four pairs of vectors, I, II, III, and IV. All vectors are of same magnitude. Magnitude of any one vector is p. In which of the pair(s) the magnitude of resultant is/are greater than p ?



Pair I Pair II Pair III Pair IV

(A) I (B) II (C) III (D) IV

30. Which of the following statement(s) is/are true about a vector quantity :

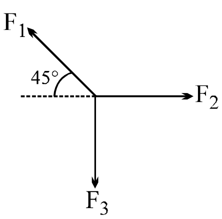
(A) A physical quantity is a vector only if it obeys law of vector addition

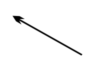
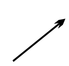
(B) Two vector quantities are equal if they have same direction and magnitude even if they represent two different physical quantities.

(C) The result of scalar product of two vectors is a scaler quantity, not another vector quantity

(D) Similar to scalar subtraction the order of the terms in vector difference affects the result.

31. Three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$ are represented as shown. Each of them is of equal magnitude.



Column I (Combination)		**Column II** (Approximate Direction)	
(A)	$\vec{F}_1+\vec{F}_2+\vec{F}_3$	(P)	
(B)	$\vec{F}_1-\vec{F}_2+\vec{F}_3$	(Q)	
(C)	$\vec{F}_1-\vec{F}_2-\vec{F}_3$	(R)	
(D)	$\vec{F}_2-\vec{F}_1-\vec{F}_3$	(S)	

32. $\vec{A}+\vec{B}=2\hat{i}$ and $\vec{A}-\vec{B}=4\hat{j}$ then what will be the angle between $\vec{A}$ and $\vec{B}$.

33. Two vectors have magnitudes 3 unit and 4 unit respectively. What should be the angle between them if the magnitude of the resultant is (a) 1 unit, (b) 5 unit and (c) 7 unit.

34. When two forces of magnitude P and Q are perpendicular to each other, their resultant is of magnitude R. When they are at an angle of 180° to each other their resultant is of magnitude $\frac{R}{\sqrt{2}}$. Find the ratio of P and Q.

35. A plane body has perpendicular axes OX and OY marked on it and is acted on by following forces

5P in the direction OY

4P in the direction OX

10P in the direction OA where A is the point (3a, 4a)

15P in the direction AB where B is the point (– a, a)

Express each force in the unit vector form & calculate the magnitude & direction of sum of the vector of these forces.

36. A vector $\vec{A}$ of length 10 units makes an angle of 60° with the vector $\vec{B}$ of length 6 units. Find the magnitude of the vector difference $\vec{A}-\vec{B}$ & the angle it makes with vector $\vec{A}$.

<u>Basic Maths</u>

37. If y represents distance and x-represent time, dimensions of $\frac{d^2y}{dx^2}$ are

(A) L^2T^{-2} (B) LT^{-1} (C) LT^{-2} (D) L^2T^{-1}

38. The position of vector of a particle is time dependent and it is given by $\vec{r}=5\hat{i}+10t\ \hat{j}$ where t is in seconds and $\vec{r}$ is in meters. Choose the correct statement(s) :

(A) velocity of particle is of magnitude 10 m/s

(B) Velocity of particle is of magnitude $\sqrt{125}\ \text{m/s}$

(C) direction of motion of particle is parallel to x-axis

(D) direction of motion of particle is perpendicular to x-axis.

39. For a particle moving in a straight line, the position of the particle at time (t) is given by

$x = t^3 - 6t^2 + 3t + 3$

what is the velocity of the particle when it's acceleration is zero ?

(A) – 9 ms^{-1} (B) – 12 ms^{-1} (C) 3 ms^{-1} (D) 42 ms^{-1}

40. A toy train is moving along a linear track. Its distance from the start of the track at time t is given by

$$x(t) = At + Bt^2 + Ct^5,$$

where A, B and C are constant.

(a) At time t_1, what is the instantaneous speed of the train ?

(b) Between time t = 0 and t = t_1, what is the average speed of the train ?

(c) Find acceleration at time t_1.

(d) Find dimensions of $\frac{B}{\sqrt{C}}$.

41. The position vector of a particle moving in x-y plane is given by $\vec{r} = (t^2-4)\hat{i}+(t-4)\hat{j}$. Find

(a) Equation of trajectory of the particle

(b) Time when it crosses x-axis and y-axis

42. Differentiate the following functions :

(i) $\sqrt{9-x^2}$ (ii) $\sqrt{\frac{1-x}{1+x}}$

43. Differentiate the following with respect to x

(i) $\frac{x^2+1}{x-1}$ (ii) $\sqrt{\sin\sqrt{x}}$ (iii) $e^{-x}\cos x$ (iv) $\frac{\sin x}{1+\cos x}$ (v) $\log_e\left(\frac{2}{x}\right)$

44. **Find the following integrals :**

(i) $\int_1^9 \frac{dx}{2\sqrt{x}}$ (ii) $\int_1^3 5x^3 dx$ (iii) $\int_1^\infty \frac{dx}{x^{1.4}}$ (iv) $\int_4^{16} \sqrt{t}\,.dt$

(v) $\int_{-5}^{-2} \frac{1}{1-x} dx$ (vi) $\int_0^1 (a+bx)^2 dx$

45. If velocity of a particle is given by $v = 3t^2 - 6t + 4$. Find its displacement from t = 0 to 3 secs.

46. Use the approximation $(1 + x)^n \approx 1 + nx$, $|x| << 1$, to find approximate value for

(a) $\sqrt{99}$ (b) $\frac{1}{1.01}$ (c) $124^{1/3}$

47. Use the small angle approximations to find approximate values for (a) sin 4° and (b) tan 5°.

48. Using 1AU (mean earth-sun distance) = 1.5×10^{11}m and parsec as distance at which 1 AU subtends an angle of 1 sec of arc, find parsec in metres.

EXERCISE-II

IITJEE QUESTIONS

Unit and Dimensions

1. If velocity [V] time [T] and force [F] are chosen as the base quantities, the dimensions of the mass will be

(A) $[FT^2V]$ (B) $[FTV^{-1}]$ (C) $[FT^{-1}V^{-1}]$ (D) $[FVT^{-1}]$

[JEE MAIN ONLINE-2021]

2. If time (t), velocity (v) and angular momentum (ℓ) are taken as the fundamental units. Then the dimension of mass (m) in terms of t, v and ℓ is : **[JEE MAIN ONLINE-2021]**

(A) $[t^{-2}\, v^{-1}\, \ell^{1}]$ (B) $[t^{-1}\, v^{1}\, \ell^{-2}]$ (C) $[t^{1}\, v^{2}\, \ell^{-1}]$ (D) $[t^{-1}\, v^{-2}\, \ell^{1}]$

3. The force is given in terms of time t and displacement x by the equation : F = A cos Bx +C sin Dt

The dimensional formula of $\frac{AD}{B}$ is : **[JEE MAIN ONLINE-2021]**

(A) $[M^2\, L^2\, T^{-3}]$ (B) $[M^1\, L^1\, T^{-2}]$ (C) $[M^0\, L\, T^{-1}]$ (D) $[M\, L^2\, T^{-3}]$

4. In Vander Waals equation $\left[P+\frac{a}{V^2}\right]$ [V – b] = RT; P is pressure, V is volume, R is universal gas constant and T is temperature. The ratio of constants $\frac{a}{b}$ is dimensionally equal to : **[JEE MAIN ONLINE-2022]**

(A) $\frac{P}{V}$ (B) $\frac{V}{P}$ (C) PV (D) PV^3

5. The SI unit of a physical quantity is pascal-second. The dimensional formula of this quantity will be

[JEE MAIN ONLINE-2022]

(A) $[ML^{-1}T^{-1}]$ (B) $[ML^{-1}T^{-2}]$ (C) $[ML^2T^{-1}]$ (D) $[M^{-1}L^3T^0]$

6. Velocity (v) and acceleration (a) in two systems of units 1 and 2 are related as $v_2 = \frac{n}{m^2}v_1$ and $a_2 = \frac{a_1}{mn}$ respectively. Here m and n are constants. The relations for distance and time in two systems respectively are : **[JEE MAIN ONLINE-2022]**

(A) $\frac{n^3}{m^3}L_1 = L_2$ and $\frac{n^2}{m}T_1 = T_2$ (B) $L_1 = \frac{n^4}{m^2}L_2$ and $T_1 = \frac{n^2}{m}T_2$

(C) $L_1 = \frac{n^2}{m}L_2$ and $T_1 = \frac{n^4}{m^2}T_2$ (D) $\frac{n^2}{m}L_1 = L_2$ and $\frac{n^4}{m^2}T_1 = T_2$

7. If the velocity of light c, universal gravitational constant G and Planck's constant h are chosen as fundamental quantities. The dimensions of mass in the new system is : **[JEE MAIN ONLINE-2023]**

(A) $[h^{1/2}c^{1/2}G^{-1/2}]$ (B) $[h^{-1/2}c^{1/2}G^{1/2}]$ (C) $[h^1c^1G^{-1}]$ (D) $[h^{1/2}c^{-1/2}G^1]$

8. The speed of a wave produced in water is given by $v = \lambda^a g^b \rho^c$. Where λ, g and ρ are wavelength of wave, acceleration due to gravity and density of water respectively. The values of a, b and c respectively, are :

[JEE MAIN ONLINE-2023]

(A) 1, –1, 0 (B) $\frac{1}{2}, \frac{1}{2}, 0$ (C) 1, 1, 0 (D) $\frac{1}{2}, 0, \frac{1}{2}$

9. In the equation $\left[X+\frac{a}{Y^2}\right][Y-b]=RT$, X is pressure, Y is volume, R is universal gas constant and T is temperature. The physical quantity equivalent to the ratio $\frac{a}{b}$ is : **[JEE MAIN ONLINE-2023]**

(A) Energy (B) Pressure gradient (C) Coefficient of viscosity (D) Impulse

10. If force (F), velocity (V) and time (T) are considered as fundamental physical quantity, then dimensional formula of density will be : **[JEE MAIN ONLINE-2023]**

(A) $F^2V^{-2}T^6$ (B) FV^4T^{-6} (C) $FV^{-2}T^2$ (D) $FV^{-4}T^{-2}$

11. Given below are two statements : **[JEE MAIN ONLINE-2023]**

Statements I : Astronomical unit (Au), Parsec (Pc) and Light year (ly) are units for measuring astronomical distances.

Statement II : Au < Parsec (Pc) < ly

In the light of the above statements choose the most appropriate answer from the options given below :

(A) Both statements I and Statements II are correct

(B) Statements I is correct but statements II is incorrect

(C) Both Statements I and Statements II are incorrect.

(D) Statements I is incorrect but Statements II is correct.

12. Consider two physical quantities A and B related to each other as $E = \frac{B-x^2}{At}$ where E, x and t have dimensions of energy, length and time respectively. The dimension of AB is : **[JEE MAIN ONLINE-2024]**

(A) $L^2M^{-1}T^1$ (B) $L^{-2}M^1T^0$ (C) $L^0M^{-1}T^1$ (D) $L^{-2}M^{-1}T^1$

13. If mass is written as $m = k\,c^P\,G^{-1/2}\,h^{1/2}$ then the value of P will be : (Constants have their usual meaning with k a dimensionless constant) **[JEE MAIN ONLINE-2024]**

(A) 1/3 (B) –1/3 (C) 2 (D) 1/2

14. A force is represented by $F = ax^2 + bt^{1/2}$ **[JEE MAIN ONLINE-2024]**

where x = distance and t = time. The dimensions of b^2/a are :

(A) $[ML^{-1}T^{-1}]$ (B) $[ML^2T^{-3}]$ (C) $[ML^3T^{-3}]$ (D) $[MLT^{-2}]$

15. The equation of state of a real gas is given by $\left(P+\frac{a}{V^2}\right)(V-b)=RT$, where P, V and T are pressure, volume and temperature respectively any R is the universal gas constant. The dimensions of $\frac{a}{b^2}$ is similar to that of **[JEE MAIN ONLINE-2024]**

(A) P (B) R (C) PV (D) RT

16. The position of a particle moving on x-axis is given by $x(t) = A\sin t + B\cos^2 t + Ct^2 + D$, where t is time. The dimension of $\frac{ABC}{D}$ is **[JEE MAIN ONLINE-2025]**

(A) L (B) L^3T^{-2} (C) L^2T^{-2} (D) L^2

17. Match **List-I** with **List-II**.

List-I	**List-II**
(a) Young's Modulus	(I) $ML^{-1}T^{-1}$
(b) Torque	(II) $ML^{-1}T^{-2}$
(c) Coefficient of Viscosity	(III) $M^{-1}L^{3}T^{-2}$
(d) Gravitational Constant	(IV) $ML^{2}T^{-2}$

Choose the correct answer from the options given below : **[JEE MAIN ONLINE-2025]**

(A) (a)-(I), (b)-(III), (c)-(II), (d)-(IV)
(B) (a)-(II), (b)-(I), (c)-(IV), (d)-(III)
(C) (a)-(IV), (b)-(II), (c)-(III), (d)-(I)
(D) (a)-(II), (b)-(IV), (c)-(I), (d)-(III)

18. The expression given below shows the variation of velocity (ν) with time (t), $\nu = At^2 + \frac{Bt}{C+t}$. The dimension of ABC is : **[JEE MAIN ONLINE-2025]**

(A) $[M^0 L^2 T^{-3}]$
(B) $[M^0 L^1 T^{-3}]$
(C) $[M^0 L^1 T^{-2}]$
(D) $[M^0 L^2 T^{-2}]$

19. The pair of physical quantities not having same dimensions is : **[JEE MAIN ONLINE-2025]**

(A) Torque and energy
(B) Surface tension and impulse
(C) Angular momentum and Planck's constant
(D) Pressure and Young's modulus

20. Given a charge q, current I and permeability of vacuum ?0. Which of the following quantity has the dimension of momentum ? **[JEE MAIN ONLINE-2025]**

(A) qI/μ_0
(B) $q\,\mu_0\,I$
(C) $q_2\,\mu_0\,I$
(D) $q\mu_0/I$

21. The equation for real gas is given by $\left(P + \frac{a}{V^2}\right)(V - b) = RT$, where P, V, T and R are the pressure, volume, temperature and gas constant, respectively. The dimension of ab^{-2} is equivalent to that of :

[JEE MAIN ONLINE-2025]

(A) Planck's constant
(B) Compressibility
(C) Strain
(D) Energy density

22. In an electromagnetic system, the quantity representing the ratio of electric flux and magnetic flux has dimension of $M^P L^Q T^R A^S$, where value of 'Q' and 'R' are **[JEE MAIN ONLINE-2025]**

(A) (3, –5)
(B) (–2, 2)
(C) (–2, 1)
(D) (1, –1)

23. The unit of $\sqrt{\frac{2I}{\epsilon_0 c}}$ is (I = intensity of an electromagnetic wave, c : speed of light)

[JEE MAIN ONLINE-2025]

(A) Vm
(B) NC
(C) Nm
(D) NC^{-1}

24. The dimension of $\sqrt{\frac{\mu_0}{\epsilon_0}}$ is equal to that of : (μ_0 = Vacuum permeability and ϵ_0 = Vacuum permittivity)

[JEE MAIN ONLINE-2025]

(A) Voltage
(B) Capacitance
(C) Inductance
(D) Resistance

25. Match **List-I** with **List-II**.

List-I	List-II
(a) Mass density	(I) $[ML^2 T^{-3}]$
(b) Impulse	(II) $[MLT^{-1}]$
(c) Power	(III) $[ML^2T^0]$
(d) Moment of inertia	(IV) $[ML^{-3} T^0]$

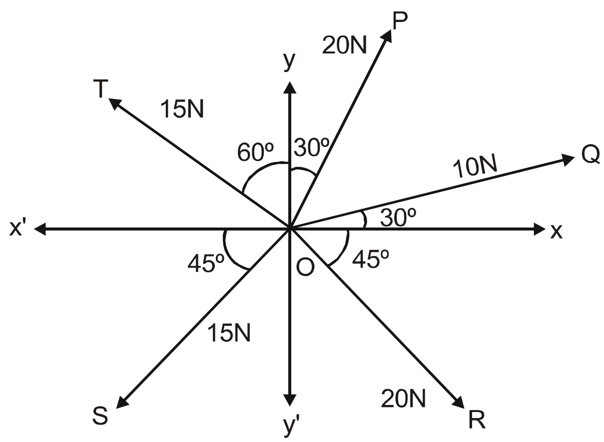
Choose the correct answer from the options given below : **[JEE MAIN ONLINE-2025]**

(A) (a)-(IV), (b)-(II), (c)-(III), (d)-(I)
(B) (a)-(I), (b)-(III), (c)-(IV), (d)-(II)
(C) (a)-(IV), (b)-(II), (c)-(I), (d)-(III)
(D) (a)-(II), (b)-(III), (c)-(IV), (d)-(I)

Vectors

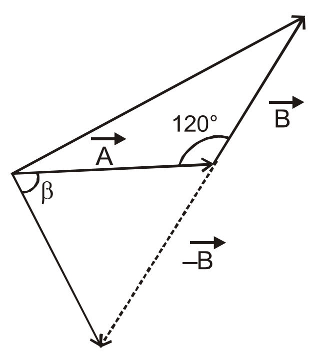
26. The resultant of these forces $\overline{OP}$, $\overline{OQ}$, $\overline{OR}$, $\overline{OS}$ and $\overline{OT}$ is approximately .... N.

[Take $\sqrt{3}-1.7$, $\sqrt{2}-1.4$. Given $\hat{i}$ and $\hat{j}$ unit vectors along x, y axis] **[JEE MAIN ONLINE-2021]**



(A) $3\hat{i}+15\hat{j}$ (B) $2.5\hat{i}-14.5\hat{j}$ (C) $9.25\hat{i}+5\hat{j}$ (D) $-1.5\hat{i}-15.5\hat{j}$

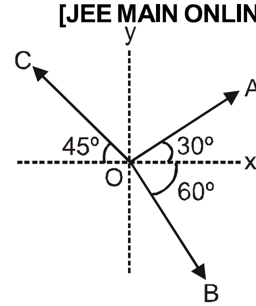
27. The angle between vector $\left(\vec{A}\right)$and$\left(\vec{A}-\vec{B}\right)$ is : **[JEE MAIN ONLINE-2021]**



(A) $\tan^{-1}\left(\dfrac{-\frac{B}{2}}{A-B\frac{\sqrt{3}}{2}}\right)$ (B) $\tan^{-1}\left(\dfrac{A}{0.7B}\right)$

(C) $\tan^{-1}\left(\dfrac{\sqrt{3}B}{2A-B}\right)$ (D) $\tan^{-1}\left(\dfrac{B\cos\theta}{A-B\sin\theta}\right)$

28. The magnitude of vectors $\overline{OA},\overline{OB}$ and $\overline{OC}$ in the given figure are equal. The direction of $\overline{OA}+\overline{OB}-\overline{OC}$ with x-axis will be : **[JEE MAIN ONLINE-2021]**

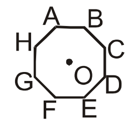


(A) $\tan^{-1}\dfrac{\left(\sqrt{3}-1+\sqrt{2}\right)}{\left(1+\sqrt{3}-\sqrt{2}\right)}$ (B) $\tan^{-1}\dfrac{\left(1+\sqrt{3}-\sqrt{2}\right)}{\left(1-\sqrt{3}-\sqrt{2}\right)}$

(C) $\tan^{-1}\dfrac{\left(1-\sqrt{3}-\sqrt{2}\right)}{\left(1+\sqrt{3}+\sqrt{2}\right)}$ (D) $\tan^{-1}\dfrac{\left(\sqrt{3}-1+\sqrt{2}\right)}{\left(1-\sqrt{3}+\sqrt{2}\right)}$

29. In an octagon ABCDEFGH of equal side, what is the sum of $\overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}+\overrightarrow{AE}+\overrightarrow{AF}+\overrightarrow{AG}+\overrightarrow{AH}$, if, $\overrightarrow{AO}=2\hat{i}+3\hat{j}-4\hat{k}$ **[JEE MAIN ONLINE-2021]**

(A) $-16\hat{i}-24\hat{j}+32\hat{k}$
(B) $16\hat{i}-24\hat{j}+32\hat{k}$
(C) $16\hat{i}+24\hat{j}-32\hat{k}$
(D) $16\hat{i}+24\hat{j}+32\hat{k}$

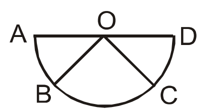


30. **Assertion A :** if A, B, C, D are four points on a semi circular arc with centre at 'O' such that

$$\left|\overrightarrow{AB}\right|=\left|\overrightarrow{BC}\right|=\left|\overrightarrow{CD}\right|, \text{ then } \overrightarrow{AB}+\overrightarrow{AC}+\overrightarrow{AD}=4\overrightarrow{AO}+\overrightarrow{OB}+\overrightarrow{OC}$$

Reason R : Polygon law of vector addition yields **[JEE MAIN ONLINE-2021]**

$$\overrightarrow{AB}+\overrightarrow{BC}+\overrightarrow{CD}=\overrightarrow{AD}=2\overrightarrow{AO}$$

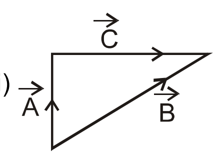
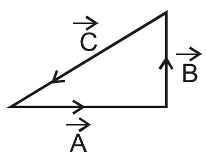
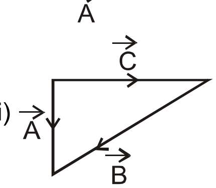
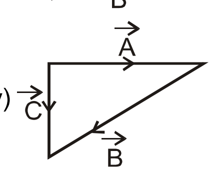


In the light of the aboves statements, choose the most appropriate answer from the options given below :
(A) A is correct but R is not correct.
(B) Both A and R are correct but R is not the correct explanation of A.
(C) Both A and R are correct and R is the correct explanation of A.
(D) A is not correct but R is correct.

31. What will be the projection of vector $\vec{A}=\hat{i}+\hat{j}+\hat{k}$ on vector $\vec{B}=\hat{i}+\hat{j}$? **[JEE MAIN ONLINE-2021]**

(A) $\sqrt{2}(\hat{i}+\hat{j})$
(B) $(\hat{i}+\hat{j})$
(C) $\sqrt{2}(\hat{i}+\hat{j}+\hat{k})$
(D) $2(\hat{i}+\hat{j}+\hat{k})$

32. Match **List I** with **List II**. **[JEE MAIN ONLINE-2021]**

List I	**List II**
(a) $\vec{C}-\vec{A}-\vec{B}=0$	(i) 
(b) $\vec{A}-\vec{C}-\vec{B}=0$	(ii) 
(c) $\vec{B}-\vec{A}-\vec{C}=0$	(iii) 
(d) $\vec{A}+\vec{B}=-\vec{C}$	(iv) 

Choose the correct answer from the options given below :
(A) (a) $\rightarrow$ (i), (b) $\rightarrow$ (iv), (c) $\rightarrow$ (ii), (d) $\rightarrow$ (iii)
(B) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (iii), (c) $\rightarrow$ (i), (d) $\rightarrow$ (ii)
(C) (a) $\rightarrow$ (iii), (b) $\rightarrow$ (ii), (c) $\rightarrow$ (iv), (d) $\rightarrow$ (i)
(D) (a) $\rightarrow$ (iv), (b) $\rightarrow$ (i), (c) $\rightarrow$ (iii), (d) $\rightarrow$ (ii)

33. Two vectors $\vec{X}$ and $\vec{Y}$ have equal magnitude. The magnitude of $(\vec{X}-\vec{Y})$ is n times the magnitude of $(\vec{X}+\vec{Y})$. The angle between $\vec{X}$ and $\vec{Y}$ is : **[JEE MAIN ONLINE-2021]**

(A) $\cos^{-1}\left(\frac{n^2+1}{n^2-1}\right)$ (B) $\cos^{-1}\left(\frac{-n^2-1}{n^2-1}\right)$ (C) $\cos^{-1}\left(\frac{n^2+1}{-n^2-1}\right)$ (D) $\cos^{-1}\left(\frac{n^2-1}{-n^2-1}\right)$

34. Two vectors $\vec{P}$ and $\vec{Q}$ have equal magnitudes. If the magnitude of $\vec{P}+\vec{Q}$ is n times the magnitude of $\vec{P}-\vec{Q}$, then angle between $\vec{P}$ and $\vec{Q}$ is : **[JEE MAIN ONLINE-2021]**

(A) $\cos^{-1}\left(\frac{n-1}{n+1}\right)$ (B) $\sin^{-1}\left(\frac{n-1}{n+1}\right)$ (C) $\cos^{-1}\left(\frac{n^2-1}{n^2+1}\right)$ (D) $\sin^{-1}\left(\frac{n^2-1}{n^2+1}\right)$

35. If the projection of $2\hat{i}+4\hat{j}-2\hat{k}$ on $\hat{i}+2\hat{j}+\alpha\hat{k}$ is zero. Then, the value of α will be **[JEE MAIN ONLINE-2022]**

36. If $\vec{A}=(2\hat{i}+3\hat{j}-\hat{k})$ m and $\vec{B}=(\hat{i}+2\hat{j}+2\hat{k})$ m. The magnitude of component of vector $\vec{A}$ along vector $\vec{B}$ will be ________ m. **[JEE MAIN ONLINE-2022]**

37. Two vectors $\vec{A}$ and $\vec{B}$ have equal magnitudes. If magnitude of $\vec{A}+\vec{B}$ is equal to two times the magnitude of $\vec{A}-\vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ will be : **[JEE MAIN ONLINE-2022]**

(A) $\sin^{-1}\left(\frac{3}{5}\right)$ (B) $\sin^{-1}\left(\frac{1}{3}\right)$ (C) $\cos^{-1}\left(\frac{3}{5}\right)$ (D) $\cos^{-1}\left(\frac{1}{3}\right)$

38. Which of the following relations is true for two unit vector $\hat{A}$ and $\hat{B}$ making an angle θ to each other?

(A) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\tan\frac{\theta}{2}$ (B) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\tan\frac{\theta}{2}$ **[JEE MAIN ONLINE-2022]**

(C) $|\hat{A}+\hat{B}|=|\hat{A}-\hat{B}|\cos\frac{\theta}{2}$ (D) $|\hat{A}-\hat{B}|=|\hat{A}+\hat{B}|\cos\frac{\theta}{2}$

39. If two vectors $\vec{P}=\hat{i}+2m\hat{j}+m\hat{k}$ and $\vec{Q}=4\hat{i}-2\hat{j}+m\hat{k}$ are perpendicular to each other. Then, the value of m will be : **[JEE MAIN ONLINE-2023]**

(A) 2 (B) 3 (C) –1 (D) 1

40. Vectors $a\hat{i}+b\hat{j}+\hat{k}$ and $2\hat{i}-3\hat{j}+4\hat{k}$ are perpendicular to each other when 3a + 2b = 7, the ratio of a to b is $\frac{x}{2}$. The value of x is ______. **[JEE MAIN ONLINE-2023]**

41. Two forces having magnitude A & A/2 are perpendicular to each other. The magnitude of their resultant is : **[JEE MAIN ONLINE-2023]**

(A) $\frac{\sqrt{5}A}{4}$ (B) $\frac{\sqrt{5}A^2}{2}$ (C) $\frac{\sqrt{5}A}{2}$ (D) $\frac{5A}{2}$

42. A vector in x–y plane makes an angle of 30° with y-axis. The magnitude of y-component of vector is $2\sqrt{3}$, The magnitude of x-component of the vector will be : **[JEE MAIN ONLINE-2023]**

(A) 6 (B) 1 (C) $\sqrt{3}$ (D) 2

43. When vector $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ is subtracted from vector $\vec{B}$, it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\vec{B}$ will be : **[JEE MAIN ONLINE-2023]**

(A) $\sqrt{6}$ (B) $\sqrt{5}$ (C) 3 (D) $\sqrt{13}$

44. If two vectors $\vec{A}$ and $\vec{B}$ having equal magnitude R are inclined at an angle θ, then **[JEE MAIN ONLINE-2024]**

(A) $|\vec{A}+\vec{B}| = 2R\sin\left(\frac{\theta}{2}\right)$ (B) $|\vec{A}+\vec{B}| = 2R\cos\left(\frac{\theta}{2}\right)$

(C) $|\vec{A}-\vec{B}| = 2R\cos\left(\frac{\theta}{2}\right)$ (D) $|\vec{A}-\vec{B}| = \sqrt{2}R\sin\left(\frac{\theta}{2}\right)$

45. A vector has magnitude same as that of $\vec{A} = (3\hat{j} + 4\hat{j})$ and is parallel to $\vec{B} = 4\hat{j} + 3\hat{j}$. The x and y components of this vector in first quadrant are x and 3 respectively where x = ________. **[JEE MAIN ONLINE-2024]**

46. Two particles are located at equal distance from origin. The position vectors of those are represented by $\vec{A} = 2\hat{i} + 3n\hat{j} + 2\hat{k}$ and $\vec{B} = 2\hat{i} - 2\hat{j} + 4p\hat{k}$, respectively. If both the vectors are at right angle to each other, the value of n^{-1} is______. **[JEE MAIN ONLINE-2025]**

Basic Maths

47. The distance of the Sun from earth is 1.5×10^{11} m and its angular diameter is (2000) s when observed from the earth. The diameter of the Sun will be : **[JEE MAIN ONLINE-2022]**

(A) 2.45×10^{10} m (B) 1.45×10^{10} m (C) 1.45×10^{9} m (D) 0.14×10^{9} m

ANSWER KEY

EXERCISE-I

Unit and Dimensions

1. D **2.** C **3.** A **4.** A **5.** B **6.** D **7.** C

8. $A \rightarrow S,\ B \rightarrow P,\ C \rightarrow Q,\ D \rightarrow R$ **9.** $a = -\frac{1}{2}$, $b = \frac{1}{2}$, $c = -\frac{1}{2}$ **10.** (a) $720\frac{\text{mi}}{\text{hr}}$ (b) $0.1\ \frac{\text{kg}}{\text{m}^2}$

11. $\therefore$ $[A] = [\sqrt{x}] = L^{+1/2}$; $[B] = ML^{1/2}T^{-2}$ **12.** $T = a\sqrt{\frac{m}{k}}$, a is dimension less constant

13. $\omega = K\sqrt{\frac{Gm}{r^3}}$ **14.** $[M] = [h^{1/2} \cdot c^{1/2} \cdot G^{-1/2}]$; $[L] = [h^{1/2} \cdot c^{-3/2} \cdot G^{1/2}]$; $[T] = [h^{1/2} \cdot c^{-5/2} \cdot G^{1/2}]$

15. $v_0 = k\sqrt{\frac{GM}{R}}$

Vectors

16. D **17.** A **18.** B,C,D **19.** B **20.** A **21.** C **22.** A **23.** D **24.** C
25. B **26.** B **27.** D **28.** D **29.** ABD **30.** ACD

31. (A) Q; (B) R; (C) P; (D) S] **32.** $\theta = 127°$ **33.** (a) 180°, (b) 90°, (c) 0 **34.** $2 \pm \sqrt{3}$

35. $5P\,\hat{j}, 4P\,\hat{i}, 6P\ \hat{i} + 8P\,\hat{j}, -12P\,\hat{i} - 9P\,\hat{j}, \sqrt{20}P, \tan^{-1}[-2]$ with the +ve x axis **36.** $2\sqrt{19}$;$\cos^{-1}\frac{7}{2\sqrt{19}}$

Basic Maths

37. C **38.** AD **39.** A

40. (a) $A + 2Bt_1 + 5Ct_1^4$. (b) $A + Bt_1 + Ct_1^4$ (c) $2B + 20Ct_1^3$ (d) $L^{1/2}\ T^{1/2}$

41. (a) $y^2 + 8y + 12 = x$; (b) crosses x axis when t = 4 sec. , crosses y axis when t = + 2 sec.

42. (i) $\frac{-x}{\sqrt{9 - x^2}}$, (ii) $\frac{-1}{(1+x)\sqrt{1-x^2}}$]

43. (i) $\frac{x^2 - 2x - 1}{(x-1)^2}$, (ii) $\frac{\cos\sqrt{x}}{4\sqrt{x\sin\sqrt{x}}}$, (iii) $-e^{-x}(\sin x + \cos x)$, (iv) $\frac{1}{1 + \cos x}$, (v) $-\frac{1}{x}$]

44. (i) 2 (ii) 100 (iii) +2.5 (iv) $\frac{112}{3}$ (v) $\ell n2$ (vi) $a^2 + ab + \frac{b^2}{3}$

45. 12 m **46.** (a) 9.95, (b) 0.99, (c) 4.987 **47.** 0.07, 0.09 **48.** $\frac{9.72}{\pi} \times 10^{16}$ m

EXERCISE-II

IITJEE MAIN QUESTIONS

Unit and Dimensions

1.	B	**2.**	D	**3.**	D	**4.**	C	**5.**	A	**6.**	A	**7.**	A
8.	B	**9.**	A	**10.**	D	**11.**	B	**12.**	A	**13.**	D	**14.**	C
15.	A	**16.**	C	**17.**	D	**18.**	A	**19.**	B	**20.**	B	**21.**	D
22.	D	**23.**	D	**24.**	D	**25.**	C						

Vectors

26.	C	**27.**	C	**28.**	C	**29.**	C	**30.**	B	**31.**	B	**32.**	B
33.	D	**34.**	C	**35.**	5	**36.**	2	**37.**	C	**38.**	B	**39.**	A
40.	1	**41.**	C	**42.**	D	**43.**	C	**44.**	B	**45.**	4	**46.**	3.00

Basic Maths

47. C